

Wetlands are cool places:

High evapotranspiration reduces surface temperature in prairie wetlands



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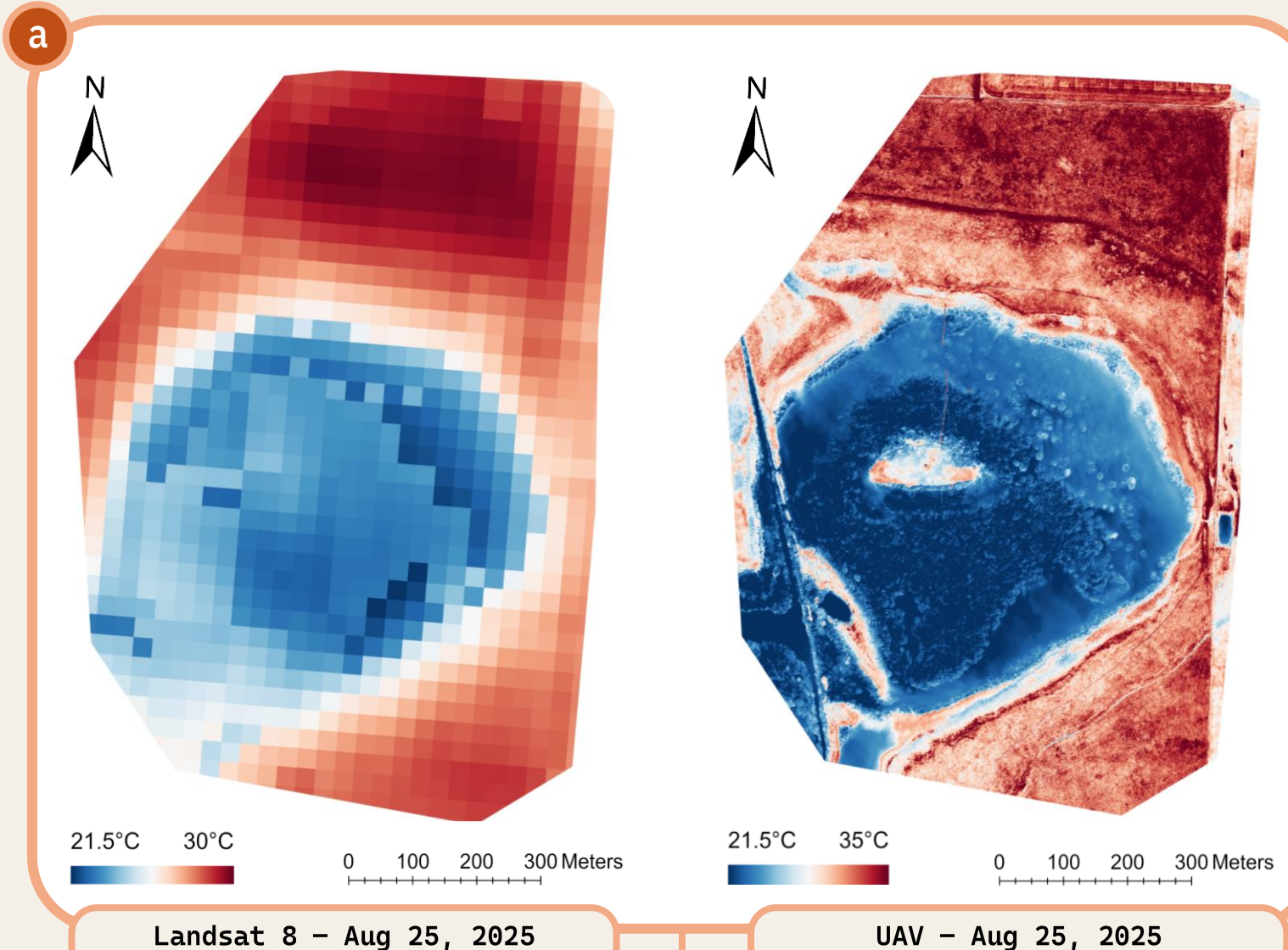
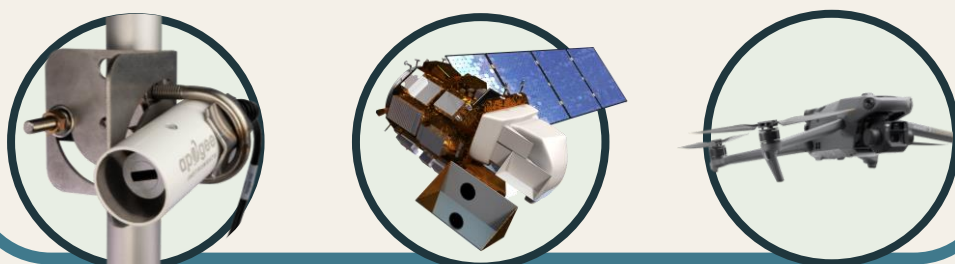
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Introduction

- Prairie pothole **wetlands provide key ecosystem services** and enhance biodiversity, carbon sequestration, and water quality
- Crucially, **wetland vegetation can also provide additional climatic regulation** via enhanced evapotranspiration, moderating sensible heat fluxes
- Altered energy balance has the potential to **provide a cooling effect on both local and global scales**
- However, **wetlands are often converted** into crop or grazing lands, **negating** both their established benefits and **potential cooling effects**
- **In this study, we analyzed the wetland cooling effect using in-situ and remote sensing methods**
- Additionally, we **compared the accuracy of each sensing method relative to in-situ** measurements

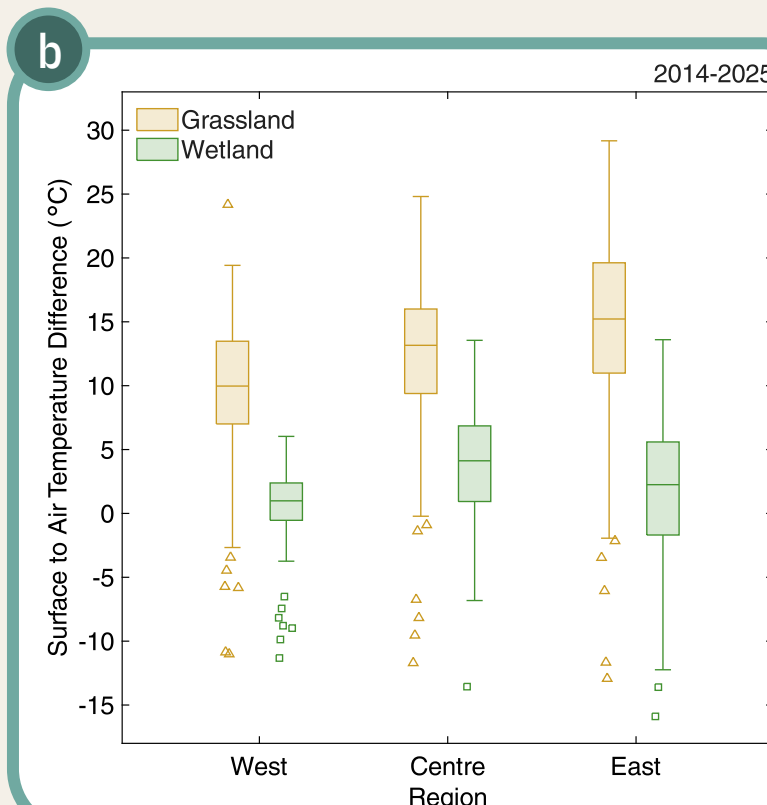
Methods

- A custom Google Earth Engine script was used to assemble **Landsat 8/9 surface temperature data for wetland and grassland sites**
- **Surface to air temperature difference was then calculated** using Alberta Climate Information Service (ACIS) air temperature data **to establish the presence of a cooling effect during 2014-2025**
- Additionally, in-situ surface/air temperature and unmanned aerial vehicle (UAV) **based surface temperature measurements were conducted** at Frank Lake, AB **to validate Landsat/ACIS data**
- A **DJI Mavic 3T Enterprise was flown coincident with a Landsat 8 overflight** on Aug 25, 2025, to enable spatial and radiometric comparisons between each remote sensing method



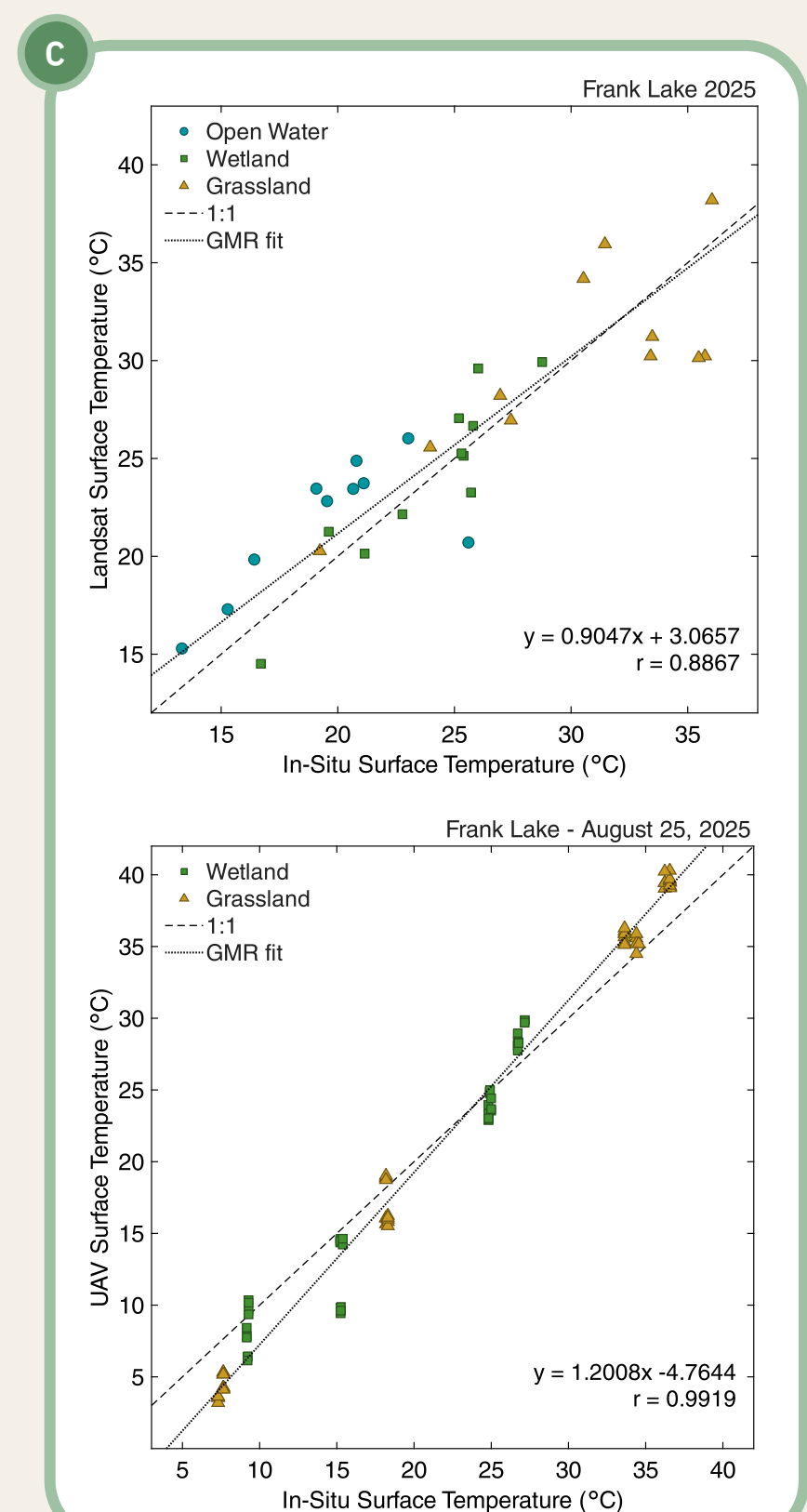
Landsat accurately portrays spatial trends within larger wetland ecosystems

UAV measurements capture extreme detail, and can be used to examine how a system warms diurnally



Wetlands provide a significant cooling effect relative to grasslands, measured across a climate gradient in Southern Alberta (mean 10.4°C, 2014-2025)

Conservation can preserve biodiversity and carbon sequestration, alongside local cooling effects, which may have impacts on the broader climate



Landsat and UAV surface temperature observations correlate with in-situ measurements, establishing the effectiveness of both methods for larger-scale investigations

Future research could analyze wetland cooling effects on a continental/planetary scale using a Landsat based workflow